MILITARY SPECIFICATION SHEET

METERS, TIME TOTALIZING, NON-HERMETICALLY SEALED, ELECTRICAL: TYPE I, 4-40V or 40-130V or 10-130V; GRADES A AND B, 2 HOLE AND 3 HOLE FLANGE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirement for procuring meters described herein shall consist of this document and the issue in effect of MIL-M-3971.

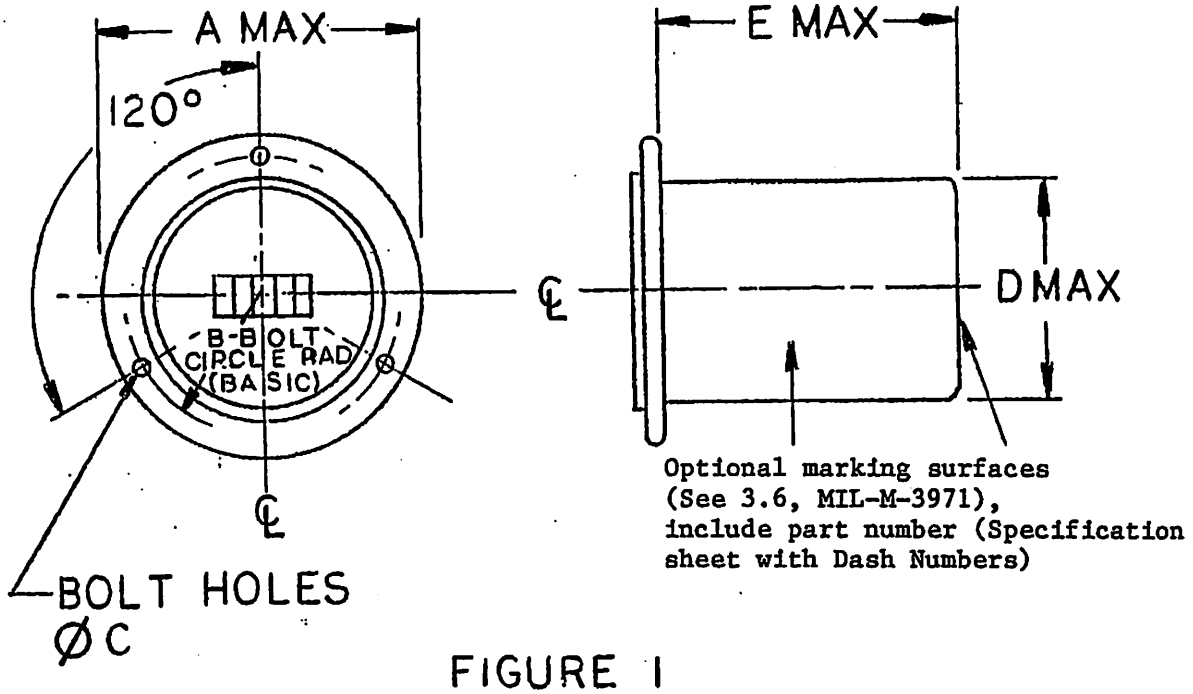


FIGURE 1

(B) denotes changes

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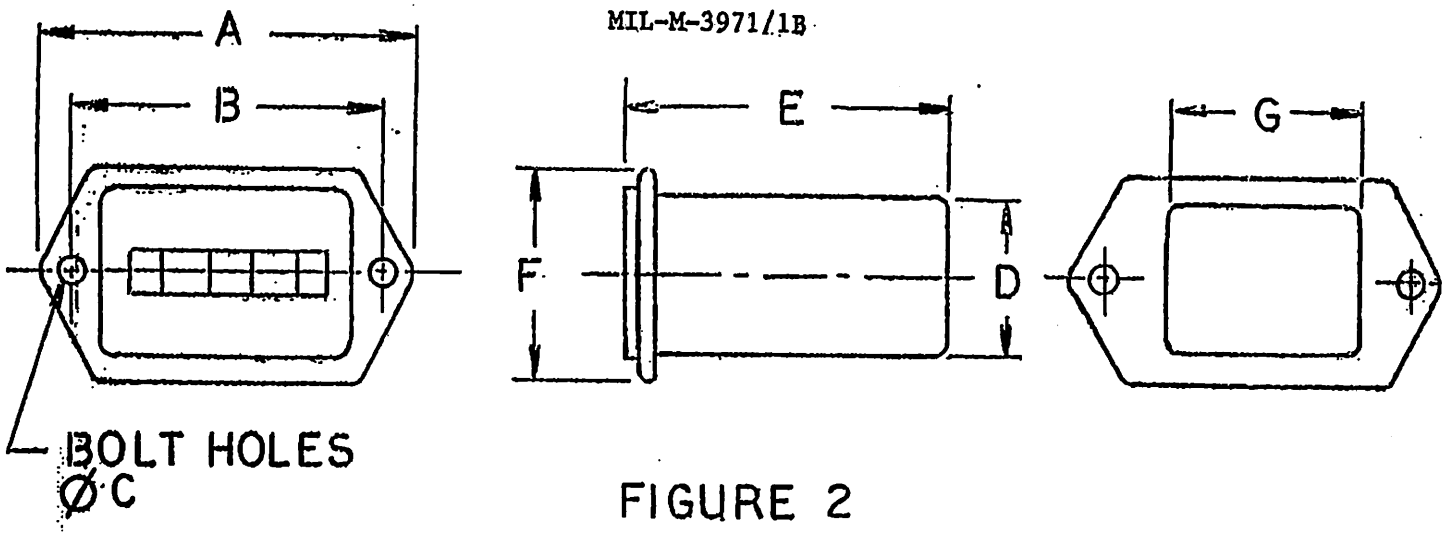


FIGURE 2

TABLE I

Dash No.	Fig	A	B	C	D	E	F	G	Grade	Voltage	Range	Readout
1	1	2.875	1.265	.150	2.210	3.00	-	-	A	4-40	10,000	9999.9
2	1	2.875	1.265	.150	2.210	3.00	-	-	A	40-130	10,000	9999.9
3	1	3.500	1.580	.150	2.625	3.00	-	-	A	4-40	10,000	9999.9
4	1	3.500	1.580	.150	2.625	3.00	-	-	A	40-130	10,000	9999.9
5	1	2.875	1.265	.150	2.210	3.00	-	-	B	4-40	10,000	9999.9
6	1	2.875	1.265	.150	2.210	3.00	-	-	B	40-130	10,000	9999.9
7	1	3.500	1.580	.150	2.625	3.00	-	-	B	4-40	10,000	9999.9
8	1	3.500	1.580	.150	2.625	3.00	-	-	B	40-130	10,000	9999.9
9	1	2.810	1.580	.150	2.135	1.75	-	-	B	10-130	10,000	9999.9
10	1	3.500	1.580	.150	2.135	1.75	-	-	B	10-130	10,000	9999.9
11	2	2.12	1.75	.150	.900	1.950	1.25	1.414	A	4-40	10,000	9999.9
12	2	2.12	1.75	.150	.900	1.950	1.25	1.414	B	4-40	10,000	9999.9

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NOTES:

1. Dimensions are in inches
2. Tolerances: Dimensional ± 0.015 inches, Bolt Holes ± .005, unless otherwise indicated . Angular ± 0.5°.
3. Range and readout are in hours and tenth of hours.

REQUIREMENTS

Dimensions of case: See Figure 1 and 2 and Table I as applicable

Colors, elapsed time indicator:

Hours: White on black background
Tenths: Black on white background

Power consumption: Not applicable

Dielectric strength:

Voltage	-	200 volts rms
Frequency	-	60 Hz
Potential	-	ac
Points of application	-	Terminals and case

Weight: 0.60 lbs maximum

Starting:

4-40V:	1.	Energize with 28V for 60 seconds at +22° (±2°C), de-energize for 180 seconds, repeat 3 times.
	2.	As above at 12V, repeat 2 times.
40-130V:	1.	As above at 120V, repeat 3 times.
	2.	As above at 85V, repeat 2 times.
10-130V:	1.	As above at 120V, repeat 3 times.
	2.	As above at 85V, repeat 2 times.
	3.	As above at 28V, repeat 3 times.
	4.	As above at 12V, repeat 2 times.

Stopping: Within 120 seconds after being de-energized.

Power supply tolerance"

4-40 voltmeters	- ½ hour at 4 volts
	- ½ hour at 40 volts
40-130 voltmeters	- ½ hour at 40 volts
	- ½ hour at 130 volts
10-130 voltmeters	- ½ hour at 10 volts
	- ½ hour at 130 volts

Part number: Consists of the basic number of this specification sheet and a dash number taken from Table I.

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Example:

M3971/1-1

Basic number
Specification sheet number
Dash number from Table I

Supersession data:

MS 16970-1 replaced by M3971/1-(5 or 6)
MS 16970-2 replaced by M3971/1-(7 Or 8)

Custodians:
Army - AR
Navy - SA
Air Force - 99

Preparing Activity:
Army - AR

Project No. 6645-0372

Review activities:
Army -
Navy - YD
Air Force - 11
DLA - GS

User activities:

Army - ER, ME
Navy - MC, AS

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